

- 3(a)(i) The statement is not correct. In order for object to come to rest, it has to experience an upward resultant force for it to decelerate to rest. Hence, force by the floor must be greater than the weight of the object.
- (ii) By conservation of momentum, the change in momentum of large man is equal and opposite to the change in momentum of small boy.

$$\Delta p_{man} = -\Delta p_{boy}$$

$$(m\Delta v)_{man} = -(m\Delta v)_{boy}$$

Since mass of large man is greater than the mass of small boy, the change in speed of small boy will be greater than the change in speed of man. Since they were both initially stationary, the small boy will move away with a larger speed than the large man. So the statement is incorrect.

(b)(i) T : tension in the rope

Apply Newton's 2nd law to the monkey which has acceleration a :

$$T - m_{monkey}g = m_{monkey}a \quad \text{-----}(1)$$

For the package to be just lifted off the ground:

$$T = m_{package}g \quad \text{-----}(2)$$

Subst. (2) into (1):

$$m_{package}g - m_{monkey}g = m_{monkey}a$$

$$15(9.81) - 10(9.81) = 10a$$

$$a = 4.9 \text{ m s}^{-2}$$

$$(b)(ii) \quad m_{package}g - T = m_{package}a$$

$$15(9.81) - T = 15a \quad \text{----} (3)$$

$$T - m_{monkey}g = m_{monkey}a$$

$$T - 10(9.81) = 10a \quad \text{----} (4)$$

$$(3) + (4) \quad 49.05 = 25a$$

$$a = 1.96 \text{ m s}^{-2}$$

$$\text{Using equation (2), } T = 10(1.96) + 10(9.81) = 118 \text{ N}$$

4(a) *(This table isn't required, just for your revision.)*